A TECHNICAL LEARNING GUIDE



SOC FUNDAMENTALS

People, Process & Security Technologies

DG INTERNS HUB

Cybersecurity Internship

Security Operations Center fundamentals: structure, roles and detection technologies

WEEK 09

PEOPLE
SOC analyst tiers

PROCESS
Monitoring to response

TECHNOLOGY
SIEM, IDS and IPS

01 — FOUNDATIONS

Introduction to the SOC

A **Security Operations Center (SOC)** is a centralized function — combining people, defined processes and supporting technology — that continuously monitors an organization's networks, systems and applications to detect, investigate and respond to cybersecurity threats. It is where raw security data is turned into decisions.

WHY ORGANIZATIONS NEED A SOC

- Attacks do not follow business hours, so monitoring must be continuous.
- Modern environments produce more logs and alerts than people can review manually.
- Early detection shortens the time an attacker remains unnoticed and limits damage.
- A consistent, documented response reduces confusion and supports compliance.

CORE OBJECTIVES

- Detect threats quickly and accurately.
- Reduce the time taken to detect and respond.
- Contain incidents and limit business impact.
- Preserve evidence and keep clear records.
- Learn from incidents to strengthen defenses.

MAJOR RESPONSIBILITIES

01 Continuous monitoring
Watching dashboards, logs and alerts across networks, endpoints and cloud services.

02 Threat detection
Spotting suspicious behavior through rules, correlation and analytics.

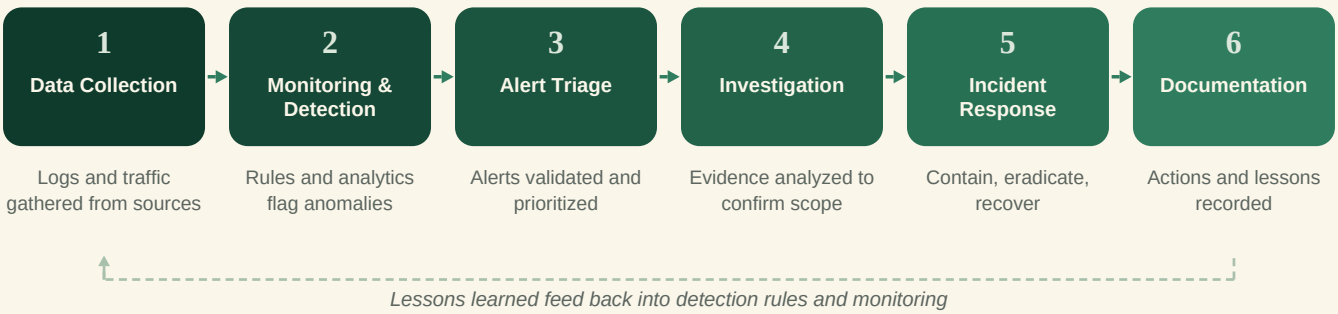
03 Alert investigation
Validating alerts and separating genuine threats from false positives.

04 Incident response
Containing, eradicating and recovering from confirmed incidents.

05 Threat intelligence
Using knowledge of attacker tactics and indicators to sharpen detection.

06 Reporting & documentation
Recording findings, actions and lessons for stakeholders and audits.

SOC WORKFLOW



REAL-WORLD EXAMPLE · SUSPICIOUS OUTBOUND TRAFFIC

A workstation begins sending steady connections to an unfamiliar external IP address at 2 a.m. Firewall and DNS logs reach the SIEM, where a rule for abnormal outbound volume raises an alert. The analyst checks the destination against threat intelligence, finds it flagged as suspicious, and reviews host logs to identify which process opened the connections. The host is isolated from the network, the case notes are completed, and the detection rule is refined afterwards.

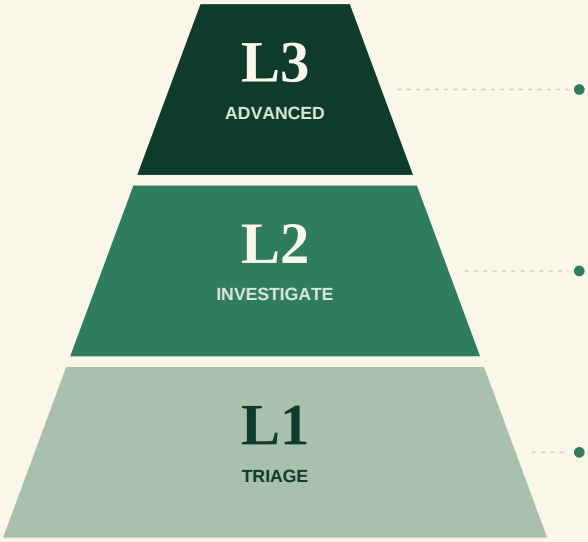
Illustrative scenario for learning purposes.

02 — PEOPLE

SOC Analyst Levels

Many SOCs organize analysts into tiers so routine alerts are handled quickly while complex incidents receive deeper expertise. The L1 / L2 / L3 model is widely used, but titles, boundaries and team sizes differ between organizations — some use fewer tiers, merge roles or add specialist teams.

THREE-LEVEL SOC HIERARCHY



L3 – Advanced Threat Analysis & Threat Hunting

Complex incident investigation · Advanced threat hunting · Malware and attacker technique analysis · Root-cause analysis · Detection engineering and security improvement

L2 – Incident Investigation & Response

Detailed alert investigation · Log and network traffic analysis · Event correlation · Incident scope and impact assessment · Containment and mitigation support

L1 – Alert Monitoring & Triage

Security dashboard monitoring · Initial alert validation · False-positive identification · Basic evidence collection · Incident escalation

ESCALATION FLOW

L1 Triage → *Needs deeper analysis* → **L2 Investigation** → *Complex or high impact* → **L3 Deep analysis**

False positive or resolved: close with notes

Findings feed new detections and playbooks

RESPONSIBILITIES AT A GLANCE

LEVEL	PRIMARY FOCUS	TYPICAL INPUTS	TYPICAL OUTCOME
L1	**Monitoring and triage of incoming alerts**	Alerts, dashboards, tickets	Closed as benign, or escalated with evidence
L2	**Investigation, correlation and response support**	Escalated cases, logs, network traffic	Confirmed scope and impact; containment actions
L3	**Advanced analysis, hunting and improvement**	Complex cases, threat intelligence, telemetry	Root cause, new detections, stronger defenses

AN ALERT MOVING BETWEEN LEVELS

L1

SIEM raises an alert: many failed logins for one account, then a success from an unfamiliar IP. L1 confirms it is not a known user or travel, adds account, timestamps and source IP to the ticket, and escalates.

L2

Reviews authentication and VPN logs, correlates the IP with other systems, and finds the account reached a file server. Supports containment by disabling the account and forcing a password reset.

L3

Studies how the credentials were obtained, hunts for the same pattern across other accounts, and builds an improved detection for repeated failures followed by a login from a new location.

Illustrative scenario. Actual escalation criteria are defined by each organization's own playbooks.

03 — TECHNOLOGY

SIEM vs IDS vs IPS

These three technologies are often mentioned together but do different jobs. IDS and IPS inspect activity for malicious patterns, while a SIEM brings together data from many sources so analysts can see the wider picture.

SIEM
Security Information and Event Management

- Centralized log collection
- Event correlation and analysis
- Alert generation
- Security monitoring and reporting

EXAMPLES

Splunk, IBM QRadar, Microsoft Sentinel, Wazuh

IDS
Intrusion Detection System

- Monitors network or host activity
- Detects suspicious traffic
- Generates alerts for analysts
- Generally passive: does not automatically block traffic

EXAMPLE

Snort in IDS mode

IPS
Intrusion Prevention System

- Detects threats and actively prevents them
- Monitors traffic inline, in the path
- Drops malicious packets or blocks connections
- Needs careful tuning to avoid blocking legitimate traffic

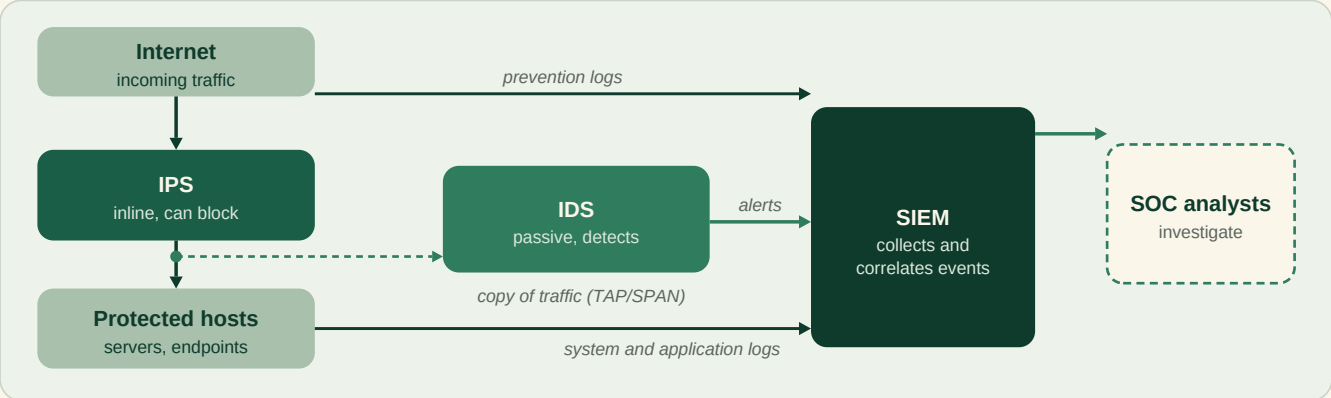
EXAMPLE

Suricata in IPS mode

COMPARISON

	SIEM	IDS	IPS
Primary purpose	Centralize, correlate and analyze security events	Detect suspicious activity and alert	Detect and block malicious activity
Data monitored	Logs and events from many sources, including IDS and IPS	Network traffic or host activity	Live network traffic passing through it
Detection capability	Correlation across sources and over time	Signature and anomaly based detection	Signature and anomaly based detection, in real time
Alert generation	Yes, from correlation rules and analytics	Yes, its primary output	Yes, plus a record of actions taken
Automatic blocking	Not by itself; response is via analysts or integrations	No, generally passive	Yes, drops or blocks traffic
Deployment approach	Agents, syslog and APIs feed a central platform	Out of band, using a TAP or SPAN copy, or on a host	Inline in the traffic path, or host based

HOW THEY WORK TOGETHER



04 — SUMMARY

Key Takeaways

SOC = PEOPLE + PROCESS + TECHNOLOGY

PEOPLE	PROCESS	TECHNOLOGY
L1, L2 and L3 analysts	Triage, escalation, documentation	SIEM, IDS and IPS

THE ESSENTIALS AT A GLANCE

L1

Monitor and triage

First line: watches alerts, validates them and escalates what matters.

L2

Investigate and respond

Digs into escalated cases, correlates evidence and supports containment.

L3

Advanced analysis and threat hunting

Handles complex incidents, hunts proactively and improves detections.

SIEM

Collect, correlate, analyze

Centralizes logs and links events from many sources into alerts.

IDS

Detect and alert

Watches traffic or hosts passively and notifies analysts.

IPS

Detect and prevent

Sits inline and blocks malicious traffic before it reaches its target.

CONCLUSION

Effective security monitoring depends on all three pillars working together. Sensors such as IDS and IPS produce and enforce detections, a SIEM brings those signals together with logs from across the environment, and tiered analysts turn them into verified findings and timely action. Understanding how each piece fits makes it easier to follow an alert from its first signal to a documented resolution — the foundation of practical security monitoring and incident response.

Learning Outcomes

- Understanding SOC structure and workflows
- Recognizing analyst responsibilities
- Differentiating SIEM, IDS and IPS
- Building foundational knowledge for practical SOC monitoring